

# SEMESTER REPORT

(Semester - 8)

## Investigating the Effectiveness of Novel Machine Learning Algorithms in Passive Acoustic Monitoring (PAM) and Species Diversity Estimation in Ensiferan Community



**Arman Singh Jaryal**

No. 2211027

*Under the guidance of*

**Dr. Rittik Deb and Dr. Subhankar Mishra**

**Evolutionary Ecology Lab**

National Institute of Science Education and Research (NISER), Bhubaneswar  
Bhubaneswar-752050, Odisha

April 2026

## Introduction

In ecology, we study how an organism interacts with its environment and other organisms. Biodiversity estimation serves as the foundation for the analysis of population and community ecology, with traditional techniques often employed to map the abundance, richness, and distribution of individuals within a habitat. The majority of species studies typically rely on active survey techniques, which require physically entering the field to estimate their presence. While camera traps are widely employed for monitoring terrestrial wildlife, they present several limitations for detailed acoustic biodiversity studies. Camera traps are often constrained by limited detection resolution for small or cryptic species, especially those that are not visually distinctive or are morphologically similar. Their ability to detect activity at night or in low-light conditions is primarily restricted to larger endothermic animals due to sensor limitations. Furthermore, camera traps typically fail to record vocalizations, nocturnal or camouflaged individuals, and species with minimal movement, thereby missing critical behavioral and ecological information. But we know that a large number of organisms use **acoustic communication** for vital functions such as attracting mates, signaling danger, and defining territory. This distinct acoustic profile allows the potential differentiation of species merely by their sounds.[1] For these reasons, our approach focuses on acoustic methods, which allow for non-invasive monitoring and identification of a broader range of vocalizing species, providing a richer dataset of behavior and presence in the community.

## Study Organism: The Ensiferan Community

Ensiferans (katydids, crickets, and wetas) are the organism of interest in this study. They form a crucial component of many terrestrial ecosystems and utilize species-specific stridulation (sound production) for communication, making bioacoustics a particularly promising survey method for them.

There is a notable lack of published research and reported data on the Biodiversity of Ensiferan community in India. The reason we must study this group is because it is considered a basal community (an older, foundational lineage in the ecosystem) that may provide key insights into evolutionary and ecological processes.[2]

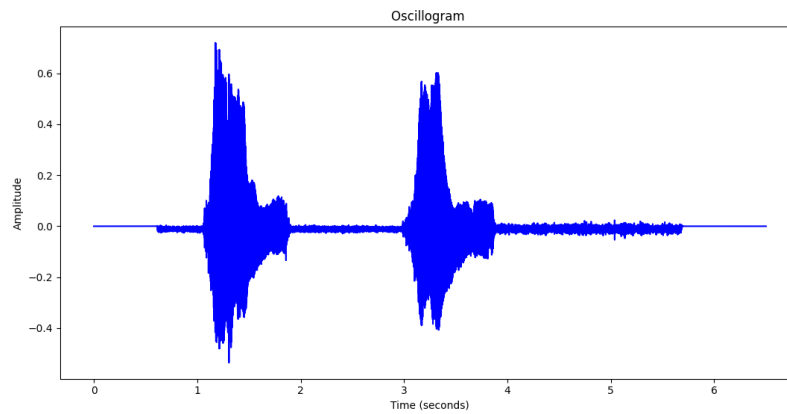
## Acoustic Monitoring: Principles and Tools

Acoustic monitoring has emerged as a powerful, non-invasive technique for surveying vocal species. However, traditional bioacoustic monitoring is fraught with challenges. The "gold standard" has historically relied on expert taxonomists to manually identify species by listening to recordings or visually inspecting their **spectrograms** (visual representations of sound). This methodology is constrained by several severe bottlenecks:

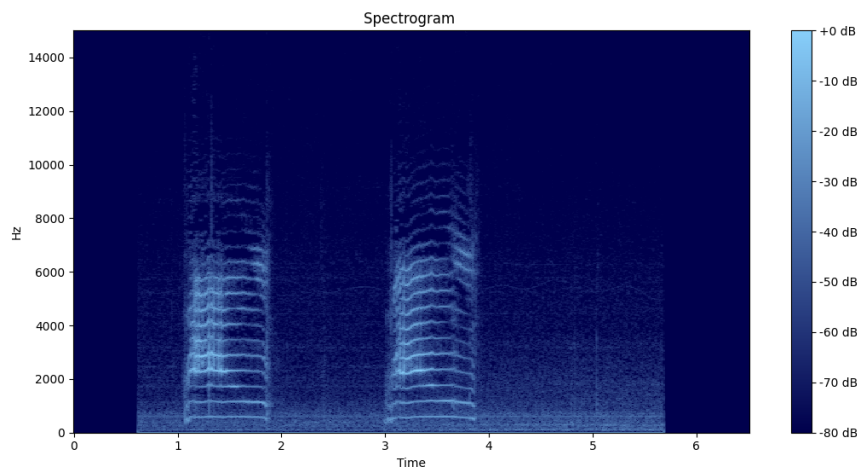
- **Time and Cost:** Manual analysis of thousands of hours of field recordings is prohibitively slow and expensive.
- **Expert Dependency:** The method's success is entirely dependent on the availability of a few, highly-trained experts, creating a "taxonomic impediment."
- **Auditory Limitations:** Many Ensiferan species, especially katydids, communicate in the **ultrasonic spectrum**, with frequencies far outside the range of human hearing, making them impossible to identify without specialized equipment and software.[3]

### Spectrograms and Oscillograms

A **spectrogram** is a visual representation of the frequency spectrum of a sound signal as it varies over time. The x-axis denotes time (in seconds), the y-axis shows frequency (in Hz), and the color intensity indicates amplitude (in dB). Spectrograms are primary tools for visually identifying species-specific call patterns and frequency structures in biological recordings. An **oscillogram** (or waveform) is the most basic graphical representation of a sound wave, displaying amplitude over time. While less informative for direct species identification than a spectrogram, oscillograms are crucial for detecting large amplitude events, evaluating recording quality, and filtering noise or silent sections in audio data. Figures 1a and 1b below illustrate these visualization techniques, using a recorded example of a cat vocalization captured with a phone microphone. The oscillogram (Fig. 1a) provides a time-domain view, while the spectrogram (Fig. 1b) offers insight into the spectral content and temporal dynamics of the call example.



(a)



(b)

Figure 1: Example of acoustic visualization tools using a cat's vocalization.(a) shows an Oscillogram (waveform) of a cat vocalization, recorded using a phone microphone. The x-axis shows time (seconds) and the y-axis shows amplitude. (b) shows Spectrogram of the same cat vocalization. The x-axis is time (seconds), the y-axis is frequency (Hz), and color intensity represents amplitude (dB). Species-specific call structure is visible as distinct frequency bands.

## The Gap in Computational Bioacoustics

To overcome these challenges, **machine learning (ML)** has been successfully applied in other bioacoustic fields. The most prominent examples are in ornithology, where models like:

- **BirdNET**
- **Merlin (from the Cornell Lab of Ornithology)**
- **PERCH (a Google research project)**

use deep neural networks to identify thousands of bird species from their calls with high accuracy. These tools have revolutionized avian monitoring, enabling large-scale, automated data collection.[4][5][6]

Despite the clear need, there are currently no robust, widely adopted, or large-scale ML models for Ensiferan identification. Although software like **Raven Pro** and **Audacity** are invaluable tools for visualizing and annotating spectrograms, they are analysis platforms, not automated identification solutions.[7][8] They do not solve the bottleneck and struggle with the real-world problem of **signal admixture** (overlapping soundscape of multiple species calling simultaneously).

## Foundation Model Selection and Core Analysis Principle

The analysis of extensive environmental audio data requires highly robust and generalizable machine learning tools. We have strategically decided to leverage NatureLM-audio as the primary foundation model for core acoustic processing tasks.

This decision is motivated by NatureLM-audio's demonstrated performance, which achieves state-of-the-art results on a wide range of bioacoustics tasks, often exceeding the capabilities of recent models that are highly specialized for their particular training data.[9] Crucially, as a large audio-language model, it exhibits powerful zero-shot generalization to unseen taxa and tasks. This feature eliminates the need for expensive and time-consuming custom model training and fine-tuning for every specific environmental context or species of interest, leading to significant savings in both time and human resource allocation. By adopting this pre-trained foundation model, we ensure a scalable, flexible, and robust analytical pipeline.

## Aim

How Effectively can Novel ML Algorithms Estimate Species Diversity in Ensiferan Community?

## Key Objectives

- To test run the model to check if the model works by using the provided test examples.
- To find any matching Ensiferan Dataset available from Nature LM's Hugging face Repository and use it to get some insight.
- To acquire data and train the model on it and find it's efficiency.
- To make complex scenario to simulate the real world challenge of identifying Richness and abundance of the species.

## NatureLM-audio: Acoustic Event Detection and Classification

The NatureLM framework processes raw audio data through a sequence of steps designed to isolate and identify relevant biological sounds. This is primarily a two-stage process:

1. **Detection (Event Localization):** The model continuously analyzes the incoming acoustic stream. The absence of an acoustic event of interest is treated as silence or background noise (no signal). When a distinct bioacoustic pattern is encountered that exceeds a learned threshold, the model performs Event Detection. This isolates the sound source (e.g., a bird call, frog vocalization, or marine mammal click) and localizes it to a precise time window within the recording. This effectively distinguishes bioacoustic signal from background noise.
2. **Classification (Event Identification):** Once an event is detected, the model uses its deep knowledge base, derived from its multimodal training, to perform Classification. This involves defining the event by identifying the species, call-type, or other contextual information associated with the localized sound. Classification is achieved through the model's ability to respond to natural language prompts (e.g., "What species is calling in this clip?"), generating a descriptive label (e.g., *Turdus migratorius* or "American Robin song").

## Data Acquisition

Acquisition of high-quality audio data is a critical component in this study. I am utilising a curated dataset provided by Souradeep Dutta, a PhD researcher in our lab, which is collected from three dry deciduous field sites Garh Jungle, Illambazar jungle and Ballavpur wildlife Sanctuary in West Bengal, as shown in Figure 1 where the white demarkated zones are the field sites.

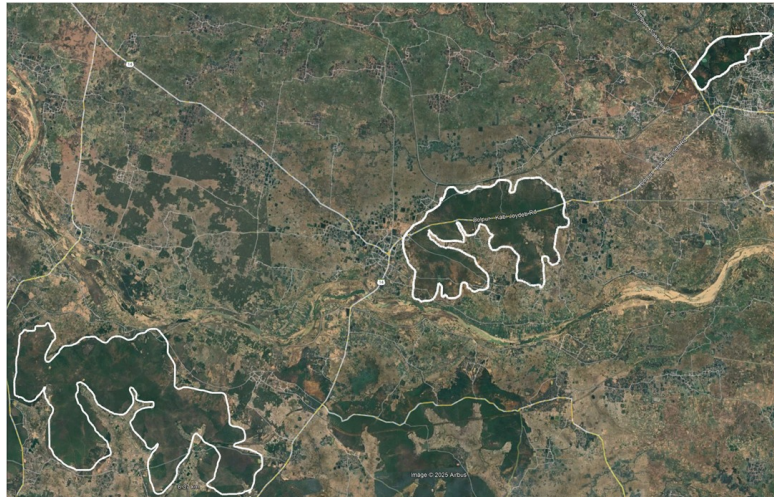


Figure 2: Satellite imagery of the three dry deciduous field sites Garh Jungle, Illambazar jungle and Ballavpur wildlife Sanctuary in West Bengal shown from left to right respectively.

This dataset comprises individual recordings of approximately 30 distinct individuals, each differing in call structure and key acoustic parameters. The majority of these recordings are approximately one minute in duration. For the purposes of model training, we will use 10-second segments extracted from these recordings, aligned with the specific requirements of the machine learning model.

Table 1: Each audio files include around 1 to 2 minute of focal recording of each species

| S.No. | Species / OTU Designation     | Number of Audio Files |
|-------|-------------------------------|-----------------------|
| 1     | <i>Bell cricket</i>           | 10                    |
| 2     | <i>Euconocephalus</i> sp.     | 5                     |
| 3     | <i>Gryllodes sigillatus</i>   | 20                    |
| 4     | <i>Gymnogryllus</i> sp.       | 9                     |
| 5     | <i>Hexacentrus</i> sp.        | 5                     |
| 6     | <i>Homeoxipha</i> sp.         | 12                    |
| 7     | <i>Mecopoda</i> sp.           | 16                    |
| 8     | <i>Modicogryllus</i> sp.      | 11                    |
| 9     | OTU 1                         | 4                     |
| 10    | OTU 10 (Type 4)               | 4                     |
| 11    | OTU 11 (Golden cricket)       | 11                    |
| 12    | OTU 12                        | 24                    |
| 13    | OTU 13 (Similar to Ballavpur) | 24                    |
| 14    | OTU 14 (March special)        | 4                     |
| 15    | OTU 15                        | 18                    |
| 16    | OTU 16 (Leaf katydid)         | 18                    |
| 17    | OTU 17                        | 1                     |
| 18    | OTU 24                        | 3                     |
| 19    | OTU 27                        | 7                     |
| 20    | OTU 2 (Flat face)             | 23                    |
| 21    | OTU 3 (Nemo long)             | 12                    |
| 22    | OTU 4 (Sand mimic)            | 3                     |
| 23    | OTU 5                         | 4                     |
| 24    | OTU 7                         | 12                    |
| 25    | OTU 8                         | 7                     |
| 26    | OTU 9 (Type 3)                | 13                    |
| 27    | <i>Satbrophylla</i> sp.       | 5                     |
| 28    | Similar to Bell               | 3                     |
| 29    | Similar to <i>Homeoxipha</i>  | 6                     |
| 30    | <i>Tarbinskelius</i> sp.      | 16                    |
| 31    | <i>Teleogryllus</i> sp.       | 14                    |
| 32    | <i>Xenogryllus</i> sp.        | 1                     |

Table 2: 10 seconds Recording split per species from 1-2 minute focal recording

| S.No. | Species / OTU Designation     | Number of Audio Files |
|-------|-------------------------------|-----------------------|
| 1     | <i>Bell cricket</i>           | 68                    |
| 2     | <i>Euconocephalus</i> sp.     | 49                    |
| 3     | <i>Gryllodes sigillatus</i>   | 56                    |
| 4     | <i>Gymnogryllus</i> sp.       | 22                    |
| 5     | <i>Hexacentrus</i> sp.        | 47                    |
| 6     | <i>Homeoxipha</i> sp.         | 124                   |
| 7     | <i>Mecopoda</i> sp.           | 164                   |
| 8     | <i>Modicogryllus</i> sp.      | 122                   |
| 9     | OTU 1                         | 51                    |
| 10    | OTU 10 (Type 4)               | 30                    |
| 11    | OTU 11 (Golden cricket)       | 71                    |
| 12    | OTU 12                        | 256                   |
| 13    | OTU 13 (Similar to Ballavpur) | 168                   |
| 14    | OTU 14 (March special)        | 22                    |
| 15    | OTU 15                        | 188                   |
| 16    | OTU 16 (Leaf katydid)         | 141                   |
| 17    | OTU 27                        | 71                    |
| 18    | OTU 2 (Flat face)             | 188                   |
| 19    | OTU 3 (Nemo long)             | 113                   |
| 20    | OTU 4 (Sand mimic)            | 19                    |
| 21    | OTU 5                         | 37                    |
| 22    | OTU 7                         | 113                   |
| 23    | OTU 8                         | 101                   |
| 24    | OTU 9 (Type 3)                | 65                    |
| 25    | <i>Sathbrophyllia</i> sp.     | 45                    |
| 26    | Similar to Bell               | 2                     |
| 27    | Similar to <i>Homeoxipha</i>  | 44                    |
| 28    | <i>Tarbinskelius</i> sp.      | 153                   |
| 29    | <i>Teleogryllus</i> sp.       | 93                    |
| 30    | <i>Xenogryllus</i> sp.        | 10                    |

## Data Processing

Training our model on long-form field recordings presented a major challenge, primarily because raw audio is rarely consistent and the abundance of the target signal can shift unpredictably throughout a single file. For instance, a typical one-minute recording might contain long stretches of silence where the animal simply stops calling, and it almost always concludes with the researcher’s voice announcing metadata like the temperature, time, and location—all of which would confuse the model if left in. To fix this, we used a custom Python script to curate the data into manageable 10-second snippets, specifically designed to capture the unique acoustic features of

each species. We didn't just rely on the script, though; we used Raven Pro to manually inspect every single clip, looking at the spectrograms and listening to the audio to ensure we were keeping only the high-quality signals and discarding anything with excessive noise or missing data. This rigorous cleaning process meant we had to narrow our scope from 30 species down to 25, as some just didn't have enough clear recordings to meet our standards, leaving us with a final set of 2,557 high-quality clips. To make the model truly resilient, we took 30 to 45 of these recordings per species and used them to simulate complex real-world scenarios, such as the loud, overlapping calls and urban noise found in disturbed city environments. By building these difficult test cases now, we can see exactly how the model holds up under pressure before we eventually transition it from the lab to actual field deployment.

Table 3: Removal of species with less Audio for testing model classification efficiency on more data per species

| S.No. | Species / OTU Designation     | Number of Audio Files |
|-------|-------------------------------|-----------------------|
| 1     | <i>Bell cricket</i>           | 68                    |
| 2     | <i>Euconocephalus</i> sp.     | 49                    |
| 3     | <i>Gryllodes sigillatus</i>   | 56                    |
| 4     | <i>Hexacentrus</i> sp.        | 47                    |
| 5     | <i>Homeoxipha</i> sp.         | 124                   |
| 6     | <i>Mecopoda</i> sp.           | 164                   |
| 7     | <i>Modicogryllus</i> sp.      | 122                   |
| 8     | OTU 1                         | 51                    |
| 9     | OTU 10 (Type 4)               | 30                    |
| 10    | OTU 11 (Golden cricket)       | 71                    |
| 11    | OTU 12                        | 256                   |
| 12    | OTU 13 (Similar to Ballavpur) | 168                   |
| 13    | OTU 15                        | 188                   |
| 14    | OTU 16 (Leaf katydid)         | 141                   |
| 15    | OTU 27                        | 71                    |
| 16    | OTU 2 (Flat face)             | 188                   |
| 17    | OTU 3 (Nemo long)             | 113                   |
| 18    | OTU 5                         | 37                    |
| 19    | OTU 7                         | 113                   |
| 20    | OTU 8                         | 101                   |
| 21    | OTU 9 (Type 3)                | 65                    |
| 22    | <i>Sathrophyllia</i> sp.      | 45                    |
| 23    | Similar to <i>Homeoxipha</i>  | 44                    |
| 24    | <i>Tarbinskius</i> sp.        | 153                   |
| 25    | <i>Teleogryllus</i> sp.       | 93                    |

# Experiments

Before diving into our full dataset, we conducted a preliminary trial run using audio from species that were already integrated into the model to establish a baseline for performance and we got to know that the model is indeed working. After that, I got the focal recordings for 32 specific species, as detailed in Table 1; these were gathered by actively approaching individual animals in the field by the lab PhDs Souradeep Dutta and Dilshad K to ensure the highest possible signal clarity. While these focal recordings were much cleaner than Passive acoustic Recording, though they still contained inevitable traces of environmental noise. We used several 10-second segments from these recordings to test whether the model's audio encoder could successfully distinguish between different individuals and species based on these short acoustic bursts. However, we recognized that relying on such clean data wouldn't give us a realistic picture of the model's ability to handle true acoustic diversity in the wild. Real-world passive acoustic monitoring is often messy and full of overlapping sounds, so we couldn't just assume success in the lab would translate to success in the field. To address this, we designed a series of artificial scenarios where we systematically added layers of complexity and noise to the data. This was done to simulate the difficult conditions of a noisy environment, ensuring a much more robust transition from controlled laboratory classification to the unpredictable reality of long-term ecological monitoring. These are the 6 different acoustic scenarios we have created:

- 1. Clean Data
- 2a. Temporal Phase Overlap
- 2b. Pair species without SNR
- 2c. Chorus Test
- 3. Pair species SNR Test
- 4. Multiple Species

## *Scenario 1: Clean Data*

Scenario 1 is the clean baseline where you have the unaltered audio first for 30 species, later denoised removed the species from the list with very noisy data which reduced it to 25 and soon to 21 species. We use it to measure the model's core species recognition ability before introducing harder acoustic conditions.

## *Scenario 2a: Temporal Phase Overlap*

Scenario 2a mixes two units of the same species across overlap levels from 0% to 100% (temporal

overlap), without adding noise. We use it to test the accuracy of the identification with varying timing overlap two same-species callers are active together.

#### *Scenario 2b: Pair Species Without SNR / Same-Species Pair with Noise Sweep*

In the implemented benchmark, Scenario 2b is two units of the same species with overlap plus SNR stress (20, 10, 5, 0 dB). We use it to quantify robustness to background noise while keeping species identity fixed, so any drop is mostly acoustic (not cross-species confusion).

#### *Scenario 2c: Chorus Test*

Scenario 2c simulates chorus behavior by stacking multiple callers from the same species (higher unit density), optionally under noise/overlap variation. We use it because real field recordings often contain many simultaneous callers of the same species, which is especially hard for counting.

#### *Scenario 3: Pair Species SNR Test*

Scenario 3 mixes one unit from species A and one unit from species B, then varies overlap and SNR. We use it to test cross-species separation under interference, which is closer to natural mixed soundscapes than single-species clips.

#### *Scenario 4: Multiple Species*

Scenario 4 is the full-complexity stress test: multi-species, multi-unit mixtures (typically 2 to 10 species with multiple units per species), with overlap and noise variation. We use it to approximate real ecological recordings where many taxa vocalize simultaneously.

## **Initial Results**

When identifying 30 distinct species from their audio recordings, the model achieved a species classification accuracy of 78.56% (5,445/6,931 correct) and a unit count accuracy of 88.00% (6,099/6,931 correct). We did a t-SNE run for clean recording to get a visual representation of how model is distinguishing each species based on their clean audio features. A t-Distributed Stochastic Neighbor Embedding (t-SNE) is a machine learning algorithm used for dimensionality reduction, specifically designed to visualize complex, high-dimensional data. Following the exclusion of noise and species (from 30 to 25 species) with data from the available pool, the model achieved a total accuracy of 98.32%, an improved performance that is clearly reflected by the dense, well-isolated clusters in the updated t-SNE visualization as mentioned in the Figure 4.

Audio embeddings are massive, complex arrays of numbers that are impossible to interpret just by looking at the raw data. t-SNE is used here as a diagnostic and validation tool. By squashing those high-dimensional acoustic signatures into a 2D scatter plot, you can visually verify if your

model is actually learning to tell the crickets apart. If the algorithm successfully maps the calls of the same species into tight, isolated groups, it proves that the acoustic features you are using are highly discriminative and your classification pipeline is working effectively.

### Scenario 1: Clean Data

For the 25-species dataset (1,049 clean recordings), Scenario 1 evaluated 1,049 samples and achieved 93.52% species accuracy (981/1,049) with 0.00% count accuracy in that run. For the 21-species dataset (1,651 recordings), Scenario 1 evaluated 1,651 samples and achieved 79.35% species accuracy (1,310/1,651) and 99.88% count accuracy (1,649/1,651). This baseline is important because it tells us whether errors are due to intrinsic class confusion or due to overlap/noise stress.

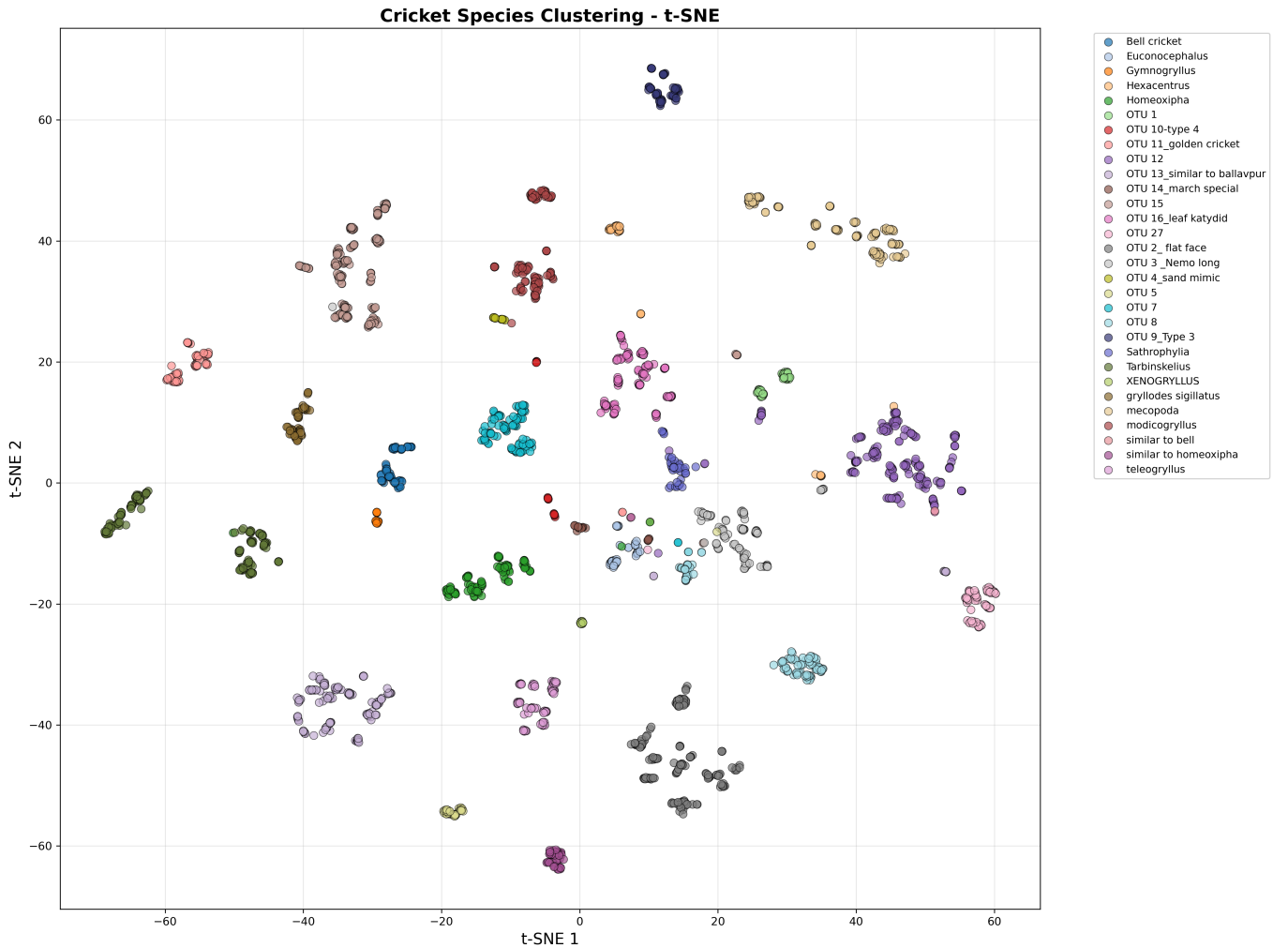


Figure 3: t-SNE visualization of optimized acoustic embeddings for 30 cricket species and OTUs.

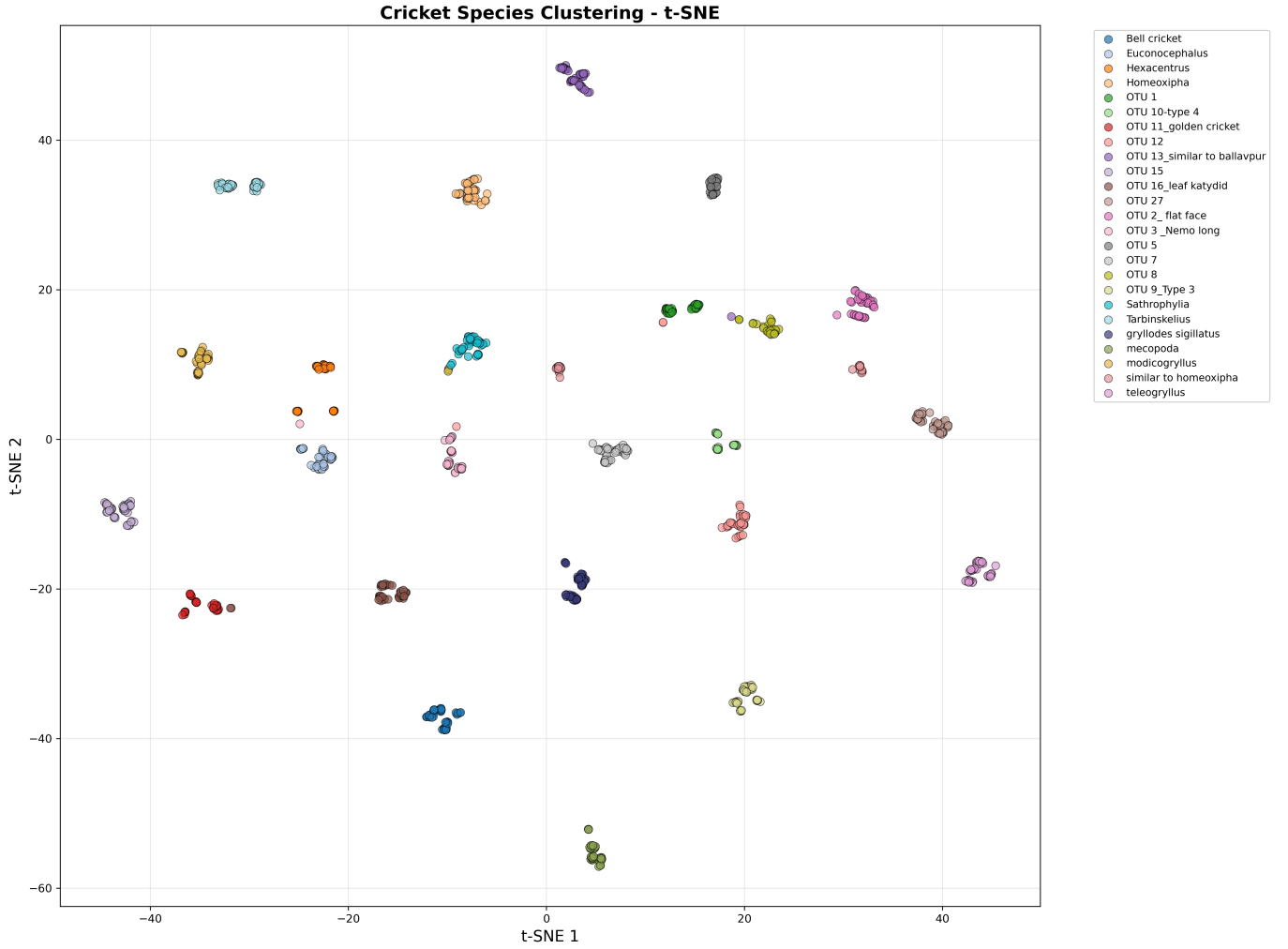


Figure 4: t-SNE visualization of acoustic features for 25 cricket species and OTUs.

Table 4: Model Performance across Scenarios

| Scenario              | Samples | Species Acc (%) | Count Acc (%) |
|-----------------------|---------|-----------------|---------------|
| <b>For 25 Species</b> |         |                 |               |
| 1                     | 1049    | 97.521449       | 99.23736892   |
| 2a                    | 50500   | 99.9980198      | 99.84950495   |
| 2b                    | 101000  | 99.98910891     | 99.24653465   |
| 2c                    | 202000  | 99.98762376     | 93.68514851   |
| 3                     | 151500  | 99.91023102     | 99.73927393   |
| 4                     | 1136250 | 99.86314631     | 99.38464246   |
| <b>For 21 Species</b> |         |                 |               |
| 1                     | 1651    | 79.345851       | 99.8788613    |
| 2a                    | 42420   | 100             | 100           |
| 2b                    | 84840   | 100             | 99.88920321   |
| 2c                    | 169680  | 99.9392975      | 96.90299387   |
| 3                     | 106050  | 99.91023102     | 99.79255068   |
| 4                     | 1136250 | 99.85020902     | 99.47837184   |

#### Scenario 2a: Temporal Phase Overlap

In the 25-species evaluation, Scenario 2a had 50,500 samples and reached 99.99% species accuracy and 99.39% count accuracy. In the 21-species evaluation, Scenario 2a had 42,420 samples and

reached 100.00% species accuracy and 100.00% count accuracy. This shows the model is extremely robust to phase/overlap effects by themselves.

#### *Scenario 2b: Pair Species Without SNR / Same-Species Pair with Noise Sweep*

For 25 species, Scenario 2b evaluated 101,000 samples with 99.99% species accuracy and 98.60% count accuracy. For 21 species, Scenario 2b evaluated 84,840 samples with 100.00% species accuracy and 99.89% count accuracy. Performance stays very high, with only small degradation in count prediction at harder SNR.

#### *Scenario 2c: Chorus Test*

In the 25-species setting, Scenario 2c had 202,000 samples, with 99.97% species accuracy and 90.07% count accuracy. In the 21-species setting, Scenario 2c had 169,680 samples, with 99.94% species accuracy and 96.90% count accuracy. Species detection remains excellent, but this scenario exposes count-estimation difficulty under dense chorus conditions.

#### *Scenario 3: Pair Species SNR Test*

In the 25-species evaluation, Scenario 3 used 151,500 samples and achieved 99.66% species accuracy and 99.27% count accuracy. In the 21-species evaluation, Scenario 3 used 106,050 samples and achieved 99.93% species accuracy and 99.79% count accuracy. These results indicate strong multi-label separation even with overlap and noise.

#### *Scenario 4: Multiple Species*

In both evaluations, Scenario 4 had 1,136,250 samples. For 25 species, it achieved 99.80% species accuracy and 98.64% count accuracy; for 21 species, it achieved 99.85% species accuracy and 99.48% count accuracy. Strong performance here shows the model scales from controlled settings to highly complex mixtures with limited degradation.

## **Future Directions**

While the current model demonstrates high classification accuracy, several methodological refinements are required for future deployment. In our current pipeline, the synthesis of Signal-to-Noise Ratio (SNR) scenarios (specifically the addition of 0, 5, 10, and 20 dB of white noise) inadvertently amplifies the underlying audio signal which is increasing the performance we need to come up with a algorithm which only increases the noise. We also observed data skew in the baseline classification results for the 21-species clean recording scenario.

To resolve this, future iterations will prioritize the curation of a strictly pristine training dataset. Currently, our clean focal recordings are processed using a simple band-pass filter; however,

this method often fails to eliminate heterospecific background calls (vocalizations from non-target species). Moving forward, we will conduct manual review and removal of these overlapping background calls to establish a truly clean acoustic baseline.

Furthermore, to maximize the utility of our limited 1-2 minute focal recordings, we plan to generate dataset pseudo-replicates. By synthesizing varying degrees of time domain within these focal recordings, we can artificially expand the dataset and improve the model's robustness.

Finally, we will deploy the optimized pipeline on a small subset of continuous, real-world acoustic data. Validating the model's efficiency against true environmental soundscapes will inform our next training phase, ultimately scaling the architecture to process and classify continuous, long-duration passive acoustic monitoring recordings.

# References

- [1] Jérôme Sueur, Almo Farina, Amandine Gasc, Nadia Pieretti, and Sandrine Pavoine. Acoustic indices for biodiversity assessment and landscape investigation. *Acta Acustica united with Acustica*, 100(4):772–781, 2014.
- [2] Jeremy Anso, Amandine Gasc, Edouard Bourguet, Laure Desutter-Grandcolas, and Hervé Jourdan. Crickets as indicators of ecological succession in tropical systems, new caledonia. *Biotropica*, 54(5):1270–1284, 2022.
- [3] Elizabeth C Blake and Ian Cross. The acoustic and auditory contexts of human behavior. *Current Anthropology*, 56(1):81–103, 2015.
- [4] Stefan Kahl, Connor M Wood, Maximilian Eibl, and Holger Klinck. Birdnet: A deep learning solution for avian diversity monitoring. *Ecological Informatics*, 61:101236, 2021.
- [5] Cornell Lab of Ornithology. Merlin bird id – free, instant bird identification help and guide for bird watchers. <https://merlin.allaboutbirds.org/>, 2021. Accessed: 2025-11-20.
- [6] Bart van Merriënboer, Vincent Dumoulin, Jenny Hamer, Lauren Harrell, Andrea Burns, and Tom Denton. Perch 2.0: The bittern lesson for bioacoustics. *arXiv preprint arXiv:2508.04665*, 2025.
- [7] Team Audacity. Audacity. *The name audacity (R) is a registered trademark of dominic mazzoni retrieved from http://audacity.sourceforge.net*, 2017.
- [8] Cornell Lab of Ornithology. Raven pro: Interactive sound analysis software (version 1.6). <https://ravensoundsoftware.com/software/raven-pro/>, 2019. Computer software.
- [9] David Robinson, Marius Miron, Masato Hagiwara, Benno Weck, Sara Keen, Milad Alizadeh, Gagan Narula, Matthieu Geist, and Olivier Pietquin. Naturelm-audio: An audio-language foundation model for bioacoustics. *arXiv preprint arXiv:2411.07186*, 2024.